Derivatives

1. Motivation

One of the important questions with functions:

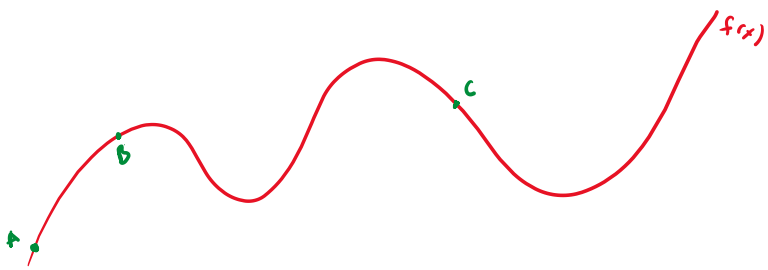
- When a function is maximized? For example: Profit function; Revenue function...
- When a function is minimized? For example: Cost function; Risk function...

We know to answer these questions when a function is "simple" (for example: quadratic function).

For more complicated functions, we will use a tool called **"Derivatives"**

2. Derivative of a function at a given point

Given a function : $y = f(x)$



(*) A derivative at a point tells us

① If the function goes up or down at that point

② How fast the function goes up / down.

(*) A derivative at a point $x = a$ is a number denoted by $f'(a)$

① If $f'(a) > 0$: $f(x)$ goes up at $x = a$ and vice versa.

② The magnitude $|f'(a)|$ is the speed of goes up / down at $x = a$

3. Calculating Derivatives

We will use several "Rules" to calculate derivatives of a function.

Notation: The derivative of $f(x)$ is denoted by

$f'(x)$ [reads: f prime of x]

<u>Rule 1</u>: Constant Rule:

If $f(x) = c$ (c is a constant)

then $f'(x) = 0$

This rule can be written as: $(c)' = 0$

<u>Rule 2</u>: Identity rule

<u>If</u>: $f(x) = x \Rightarrow f'(x) = 1$

<u>OR</u>: $(x)' = 1$

<u>Rule 3</u>: Power Rule.

If $f(x) = x^n$ (power function)

then $f'(x) = n \cdot x^{n-1}$

<u>OR</u>: $(x^n)' = n \cdot x^{n-1}$

<u>Example</u>: Find $f'(x)$

① $f(x) = x^{2026} \Rightarrow f'(x) = 2026 \cdot x^{2026-1}$

$\Rightarrow f'(x) = 2026\, x^{2025}$

② $f(x) = x^{10} \Rightarrow f'(x) = 10 \cdot x^{9}$

③ $f(x) = x^{2} \Rightarrow f'(x) = 2 \cdot x^{1} = 2x$

④ $f(x) = x^{2.6} \Rightarrow f'(x) = 2.6 * x^{1.6}$

⑤ $f(x) = \dfrac{1}{x^3}$

$f(x) = \dfrac{1}{x^3} = x^{-3}$

$\Rightarrow f'(x) = -3 \cdot x^{-3-1} = -3x^{-4}$

$= \dfrac{-3}{x^4}$

$$\boxed{\text{Notice: } \frac{1}{x^k} = x^{-k}}$$

⑥ $f(x) = \sqrt[3]{x} = x^{1/3}$

$f'(x) = \frac{1}{3} \cdot x^{1/3 - 1} = \frac{1}{3} x^{-2/3}$

Practice: Find $f'(x)$

① $f(x) = 2026$

② $f(x) = x$

③ $f(x) = x^7$

④ $f(x) = x^{20}$

⑤ $f(x) = x^{7/3}$

⑥ $f(x) = x^{16}$

⑦ $f(x) = \frac{1}{x^6}$

⑧ $f(x) = \frac{1}{x^9}$

⑨ $f(x) = \frac{1}{x^{20}}$

Rule 4: Scalar multiplication rule.

$$(k \cdot f(x))' = k \cdot f'(x)$$

Example:

① $f(x) = 6x^3 \Rightarrow f'(x) = 6 \cdot 3 \cdot x^2$

$= 18x^2$

② $h(x) = 20x^9 \Rightarrow h'(x) = 20 \cdot 9x^8$

$= 180x^8$

Rule 5 : Addition | Subtraction Rules

$$[h(x) + g(x)]' = h'(x) + g'(x)$$

$$[h(x) - g(x)]' = h'(x) - g'(x)$$

Example :

$$(x^3 + x^4)' = (x^3)' + (x^4)'$$

$$= 3x^2 + 4x^3$$

$$(x^9 - 21)' = (x^9)' - (21)'$$

$$= 9x^8$$

Rule 6 : Product Rule

$$[f(x) \cdot s(x)]' = f'(x) \cdot s(x) + f(x) \cdot s'(x)$$

Example :

$$h(x) = (x^2 + 1) \cdot (x^3 - 2)$$

$$\Rightarrow h'(x) = [(x^2+1) \cdot (x^3-2)]'$$

$$= (x^2+1)' \cdot (x^3-2) + (x^2+1) \cdot (x^3-2)'$$

$$= 2x(x^3-2) + (x^2+1)\cdot(3x^2)$$

<u>Practice</u>: Find $f'(x)$. Is $f(x)$ going up or down when $x = 1$?

① $f(x) = 9x^3$

$\Rightarrow f'(x) = 9 \cdot 3 \cdot x^2 = 27x^2$

$f'(1) = 27 > 0 \Rightarrow f(x)$ is going up when $x = 1$

② $f(x) = 6x^7$

③ $f(x) = 6x^3 - 14x^2$

④ $f(x) = 9x^3 + 10x + 7$

⑤ $f(x) = (x^2+x)\cdot(x^2-4x)$

Rule 7: Quotient Rule.

$$\left(\frac{\text{Top}}{\text{Bot}}\right)' = \frac{(\text{Top})' \cdot \text{Bot} - (\text{Top}) \cdot (\text{Bot})'}{(\text{Bot})^2}$$

Example:

Find $f'(x)$. Is $f(x)$ going up or down when $x = 2$?

① $$f(x) = \frac{x^2 + x + 1}{x^2 - 6x}$$

$$\Rightarrow f'(x) = \frac{(x^2+x+1)' \cdot (x^2 - 6x) - (x^2+x+1) \cdot (x^2 - 6x)'}{(x^2 - 6x)^2}$$

$$= \frac{(2x+1) \cdot (x^2 - 6x) - (x^2+x+1) \cdot (2x - 6)}{(x^2 - 6x)^2}$$

$$f'(2) = \frac{5 \cdot (-8) - (7) \cdot (-2)}{2} < 0$$

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$\Rightarrow$ $f(x)$ goes down when $x = 2$

<u>Practice</u> : Find $f'(x)$

① $f(x) = \dfrac{x+1}{x+2}$

② $f(x) = \dfrac{x^3 - 3}{x^3 + 3}$

$$\left(\frac{\text{Top}}{\text{Bot}}\right)' = \frac{(\text{Top})' \cdot \text{Bot} - (\text{Top}) \cdot (\text{Bot})'}{(\text{Bot})^2}$$

$$= 2x(x^3-2) + (x^2+1)\cdot(3x^2)$$

<u>Practice</u>: Find $f'(x)$. Is $f(x)$ going up or down when $x = 1$?

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